

LAB

QUESTION: 539

What protocol allows an engineer to back up 20 network router configurations globally while using the copy function?

- ☐ A TCP
- ☐ B SMTP
- ☒ C FTP
- ☐ D SNMP

Answer(s): C

```
Total doors: 0
Appl doors: 0
Normal doors: 0
Queued Packets: 0
```

Refer to the exhibit. What is the next step to complete the implementation for the partial NAT configuration shown?

- A. Modify the access list for the internal network on e0/1.
- B. Reconfigure the static NAT entries that overlap the NAT pool.
- C. Apply the ACL to the pool configuration.
- ☒ D. Configure the NAT outside interface. **Most Voted**



EXAM 200-301 TOPIC 1 QUESTION 661 DISCUSSION

Actual exam question from Cisco's 200-301

Question #: 661

Topic #: 1

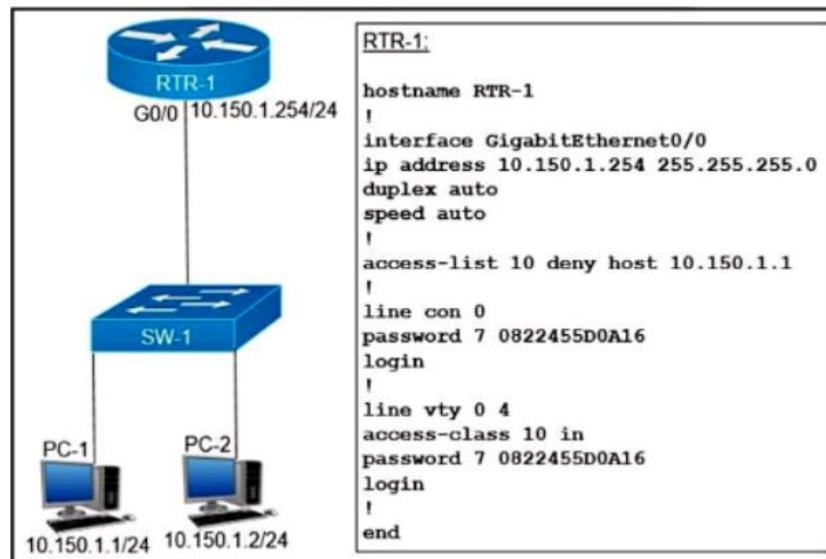
[\[All 200-301 Questions\]](#)

Which action implements physical access control as part of the security program of an organization?

- A. setting up IP cameras to monitor key infrastructure
- B. configuring a password for the console port
- C. backing up syslogs at a remote location
- D. configuring enable passwords on network devices

QUESTION: 664

Refer to the exhibit. An access list is created to deny Telnet access from host PC-1 to RTR-1 and allow access from all other hosts. A Telnet attempt from PC-2 gives this message: "% Connection refused by remote host." Without allowing Telnet access from PC-1, which action must be taken to permit the traffic?



- ☒ A Add the access-list 10 permit any command to the configuration.
- ☐ B Remove the access-class 10 in command from line vty 0 4
- ☐ C Add the ip access-group 10 out command to interface g0/0.
- ☐ D Remove the password command from line vty 0 4.

Question #905

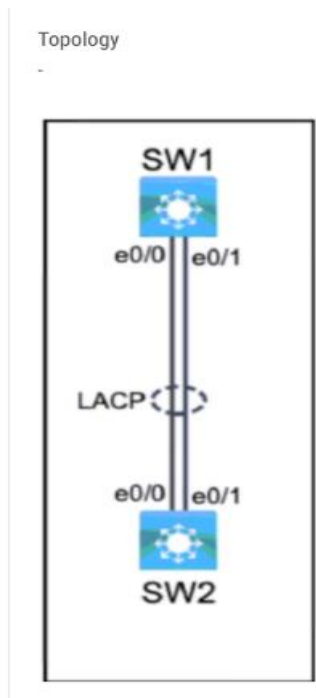
Topic 1

DRAG DROP

Drag and drop the device behaviors from the left onto the matching HSRP state on the right.

has heard from the neighbor device and is receiving hello packets	Active
is forwarding packets	Learn
is ready to forward packets if the device that is currently forwarding packets fails	Listen
is transmitting and receiving hello packets	Speak
is waiting to hear from the neighbor device	Standby

Dạng 1:



Tasks

Physical connectivity is implemented between the two Layer 2 switches, and the network connectivity between them must be configured.

1. Configure an LACP EtherChannel and number it as 44; configure it between switches SW1 and SW2 using interfaces Ethernet0/0 and

Ethernet0/1 on both sides. The LACP mode must match on both ends.

2. Configure the EtherChannel as a trunk link.

3. Configure the trunk link with 802.1q tags.

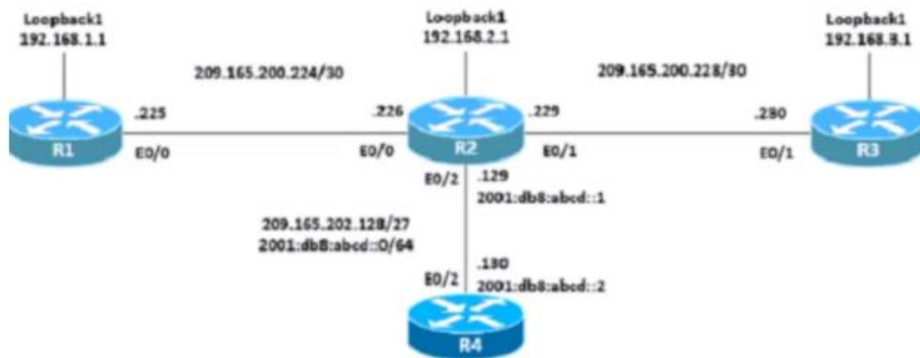
4. Configure VLAN 'MONITORING' as the untagged VLAN of the EtherChannel.(**Show VLAN**)

Đáp án:

S1	S2
<pre>Int range e0/0-1 Shut Channel-group 44 mode active No shut Int port-channel 44 Switchport trunk encapsulation dot1q Switchport mode trunk Switchport trunk native Vlan 746 Copy running-config startup-config</pre>	<pre>Int range e0/0-1 Shut Channel-group 44 mode active No shut Int port-channel 44 Switchport trunk encapsulation dot1q Switchport mode trunk Switchport trunk native Vlan 746 Copy running-config startup-config</pre>

Dạng 2:

Topology



Task:

Connectivity between four routers has been established. IP connectivity must be configured in the order presented to complete the implementation. No dynamic routing protocols are included.

1. Configure static routing using host routes to establish connectivity from router R3 to the router R1 Loopback address using the source IP of 209.165.200.230.
2. Configure an IPv4 default route on router R2 destined for router R4.
3. Configure an IPv6 default router on router R2 destined for router R4.

Task 1:

R3(config)#ip route 192.168.1.1 255.255.255.255 209.165.200.229 (R3→loopback R1)

R2(config)#ip route 192.168.1.1 255.255.255.255 209.165.200.225 (R2→loopback R1)

R1(config)# ip route 209.165.200.230 255.255.255.255 209.165.200.226 (R1→R3)

Task 2:

R2(config)#ip route 0.0.0.0 0.0.0.0 209.165.202.130

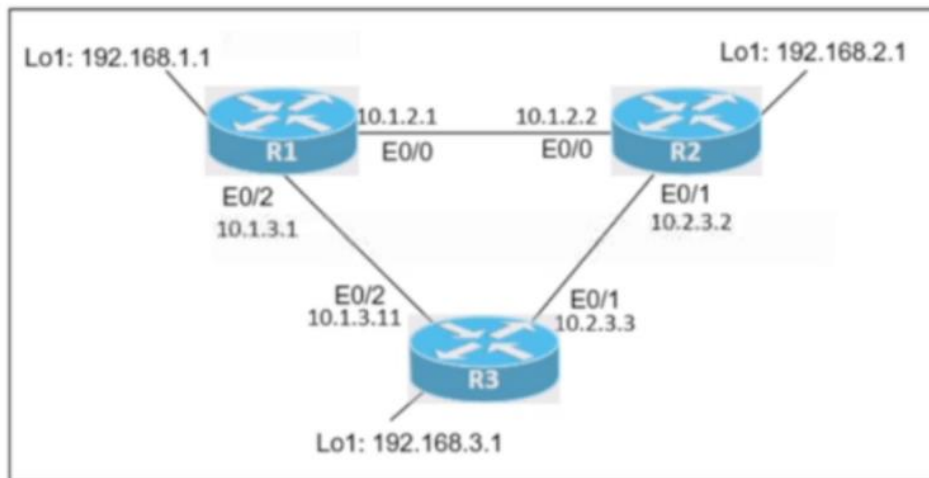
Task 3:

R2(config)#ipv6 unicast-routing

R2(config)#ipv6 route ::/0 2001:db8:abcd::2

Dạng 3:

Topology



- Task:

Connectivity between three routers has been established, and IP services must be configured in the order presented to complete the

implementation. Tasks assigned include configuration of NAT, NTP, DHCP, and SSH services.

1. All traffic sent from R3 to the R1 Loopback address must be configured for NAT on R2. All source addresses must be translated from R3 to the IP address of Ethernet0/0 on R2, while using only a standard access list named PUBNET. To verify, a ping must be successful to the R1 Loopback address sourced from R3. Do not use NVRAM NAT configuration.

R2: (NAT hết địa chỉ của R3)

Ip access-list standard PUBNET

Permit host 10.2.3.3

Permit host 10.1.3.11

Permit host 192.168.3.1

Exit

Int e0/1

Ip nat inside

Int e0/0

Ip nat outside

Ex

Ip nat inside source list PUBNET int e0/0 overload

2. Configure R1 as an NTP server and R2 as a client, not as a peer, using the IP address of the R1 Ethernet0/2 interface. Set the clock on the NTP server for midnight on May 1, 2018.

R1:

Clock set 00:00:00 May 1 2018

Ntp master 1

R2:

Ntp server 10.1.3.1

3. Configure R1 as a DHCP server for the network 10.1.3.0/24 in a pool named NETPOOL. Using a single command, exclude addresses 1 - 10 from the range. Interface Ethernet0/2 on R3 must be issued the IP address of 10.1.3.11 via DHCP.

R1:

ip dhcp excluded-address 10.1.3.1 10.1.3.10

ip dhcp pool NETPOOL

network 10.1.3.0 255.255.255.0

R3:

Interface e0/2

Ip address dhcp

4. Configure SSH connectivity from R1 to R3, while excluding access via other remote connection protocols. Access for user netadmin and password N3t4ccess must be set on router R3 using RSA and 1024 bits. Verify connectivity using an SSH session from router R1 using a destination address of 10.1.3.11. Do NOT modify console.

R3:

Ip domain-name ccna.com

Crypto key generate rsa

General-keys modulus 1024

Username netadmin secret N3t4ccess

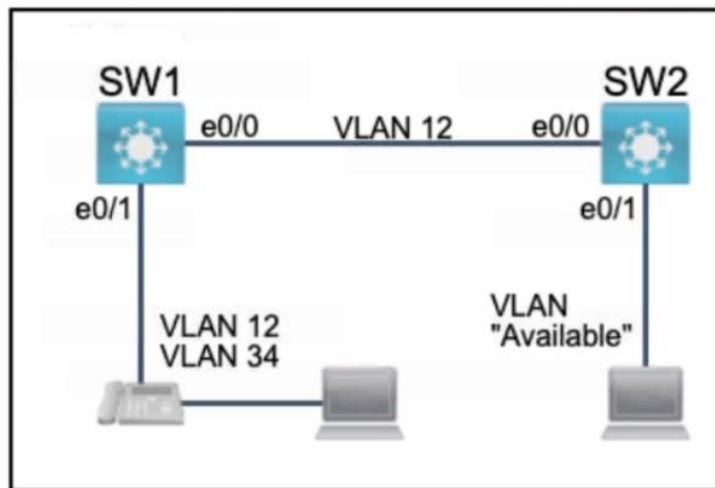
Line vty 0 15

Transport input ssh

Login local

Nhớ lưu cấu hình

Dạng 4:



Tasks

All physical cabling between the two switches is installed. Configure the network connectivity between the switches using the designated VLANs

and interfaces.

1. Configure VLAN 12 named Compute and VLAN 34 named Telephony where required for each task.
2. Configure Ethernet0/1 on SW2 to use the existing VLAN named Available.

Show VLAN → Got VLAN-ID of the Vlan Available → Add e0/1 to vlan Available

3. Configure the connection between the switches using access ports.

SW1(config)#interface e0/0

SW1(config-if)#switchport mode access

SW1(config-if)#switchport access vlan 12

SW2(config)#interface e0/0

SW2(config-if)#switchport mode access

SW2(config-if)#switchport access vlan 12

4. Configure Ethernet0/1 on SW1 using data and voice VLANs.

SW1(config)#interface e0/1

SW1(config-if)#switchport mode access

SW1(config-if)#switchport voice vlan 34

SW1(config-if)#switchport access vlan 12

5. Configure Ethernet0/1 on SW2 so that the Cisco proprietary neighbor discovery protocol is turned off for the designated interface only.

Task 5

SW2(config)#cdp run

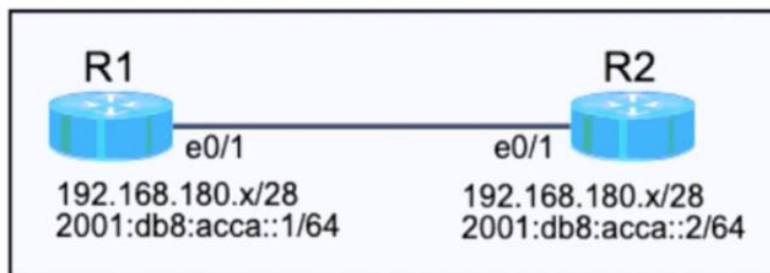
SW2(config)#interface gigabitEthernet 0/1

SW2(config-if)#no cdp enable

Nhớ lưu cấu hình

Dạng 5:

Topology



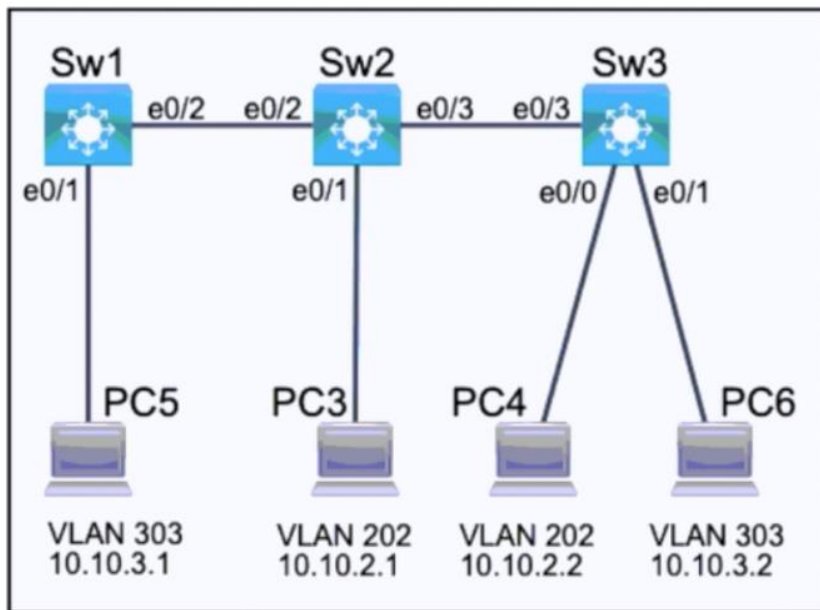
Tasks

ipv6 unicast-routing

Configure IPv4 and IPv6 connectivity between two routers. For IPv4, use a /28 network from the 192.168.180.0/24 private range. For IPv6, use the 9rst /64 subnet from the 2001:0db8:acca::/48 subnet.

1. Using Ethernet0/1 on routers R1 and R2, configure the next usable /28 from the 192.168.180.0/24 range. The network 192.168.180.0/28 is unavailable.
2. For the IPv4 /28 subnet, router R1 must be configured with the 9rst usable host address.
(17)
3. For the IPv4 /28 subnet, router R2 must be configured with the last usable host address.
(30)
4. For the IPv6 /64 subnet, configure the routers with the IP addressing provided from the topology.
5. A ping must work between the routers on the IPv4 and IPv6 address ranges.

Dạng 6:



Tasks

Three switches must be configured for Layer 2 connectivity. The company requires only the designated VLANs to be configured on their respective

switches and permitted across any links between switches for security purposes. Do not modify or delete VTP configurations.

The network needs two user-defined VLANs configured:

VLAN 202: MARKETING

VLAN 303: FINANCE

1. Configure the VLANs on the designated switches and assign them as access ports to the interfaces connected to the PCs.
2. Configure the e0/2 interfaces on Sw1 and Sw2 as 802.1q trunks with only the required VLANs permitted.
3. Configure the e0/3 interfaces on Sw2 and Sw3 as 802.1q trunks with only the required VLANs permitted.

SW1:

interface e0/2

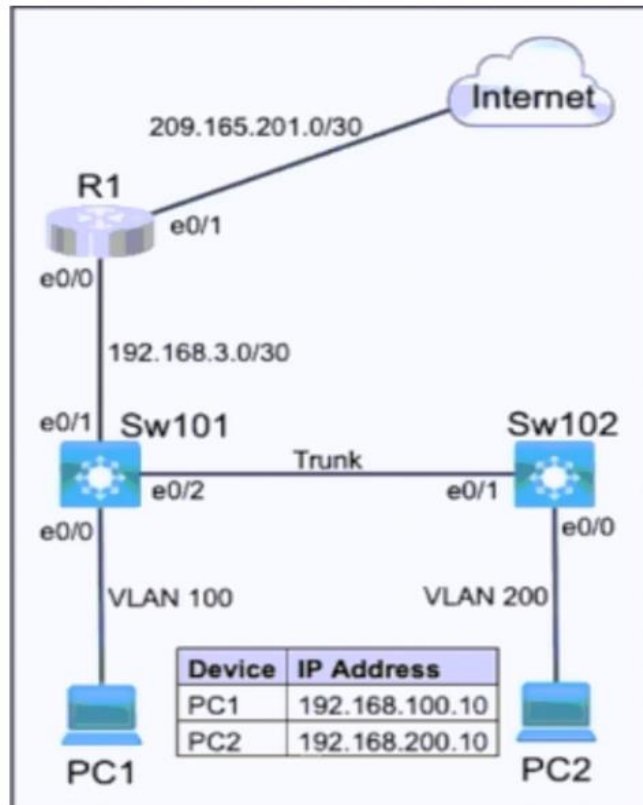
switchport trunk encapsulation dot1q

switchport mode trunk

switchport trunk allowed vlan 303

Nhớ lưu cấu hình

Dạng 7:



Task 1

Configure a local account on Sw101 with telnet access only on virtual ports 0-4. Use the following information:

- Username: support
- Password: max2learn
- Privilege level: Exec mode

```
sw101(config)#username support privilege 15 password max2learn
```

```
sw101(config)#line vty 0 4
```

```
sw101(config-line)#transport input telnet
```

```
sw101(config-line)#login local
```

Task 2

Configure and apply a single NACL on Sw101 using the following:

Name: ENT_ACL

Restrict only PC2 on VLAN 200 from pinging PC1

Allow only PC2 on VLAN 200 to telnet to Sw101

Prevent all other devices from telnetting from VLAN 200

Allow all other network traffic from VLAN 200

```
ip access-list extended ENT_ACL
```

```
deny icmp host 192.168.200.10 host 192.168.100.10
```

```
permit tcp host 192.168.200.10 any eq telnet
```

```
deny tcp any any eq telnet
```

```
permit ip any any
```

```
interface Vlan 200
```

```
ip access-group ENT_ACL in
```

```
line vty 0 4
```

```
access-class ENT_ACL in
```

Task 3

Configure security on interface Ethernet 0/0 of Sw102:

Set the maximum number of secure MAC addresses to four.

Drop packets with unknown source addresses until the number of secure MAC addresses drops below the configured maximum value. No notification action is required.

Allow secure MAC addresses to be learned dynamically.

```
sw102(config)#interface Ethernet 0/0
```

```
sw102(config-if)#switchport port-security
```

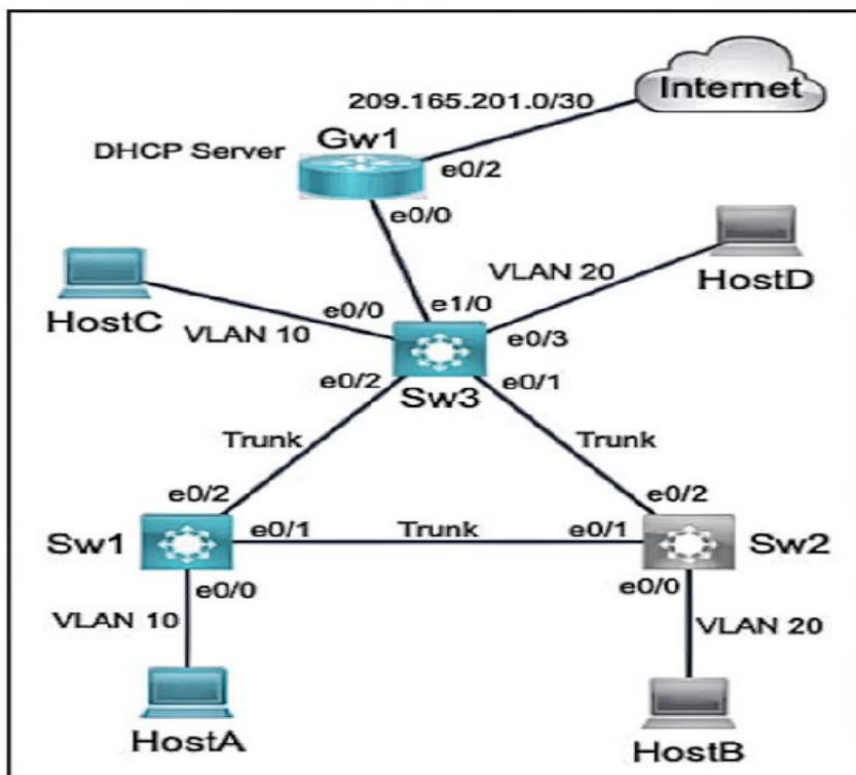
```
sw102(config-if)#switchport port-security maximum 4
```

```
sw102(config-if)#switchport port-security violation protect
```

```
sw102(config-if)#switchport port-security mac-address sticky
```

Nhớ lưu cấu hình

Dạng 8:



Tasks

Refer to the topology. All physical cabling is in place. Configure local users accounts, modify the Named ACL (NACL), and configure DHCP Snooping. The current contents of the NACL must remain intact.

Task 1

Configure a local account on Gw1 with telnet access only on virtual ports 0-4. Use the following information:

- Username: wheel
- Password: lock3path
- Algorithm type: Scrypt
- Privilege level: Exec mode

GW1:

```
R1(config)#username wheel privilege 15 algorithm-type scrypt secret lock3path
```

```
line vty 0 4
```

```
transport input telnet
```

```
login local
```

Task 2

-

Configure and apply a NACL on Gw1 to control network traffic from VLAN 10:

- Name: CORP_ACL
- Allow BOOTP and HTTPS
- Restrict all other traffic and log the ingress interface, source MAC address, the packet's source and destination IP addresses, and ports

Task 2

GW1(config)#ip access-list extended CORP_ACL

GW1(config-ext-nacl)#permit udp any any eq 68,67

GW1(config-ext-nacl)#permit tcp any any eq 443

GW1(config-ext-nacl)#deny ip any any log-input

GW1(config)#interface gigabitEthernet 0/0.10

GW1(config-subif)#ip access-group CORP_ACL in

Task 3

Configure Sw1:

- Enable DHCP Snooping for VLAN 10
- Disable DHCP Option-82 data insertion
- Enable DHCP Snooping MAC address verification
- Enable trusted interfaces

Task 3

SW1(config)#ip dhcp snooping vlan 10

SW1(config)#no ip dhcp snooping information option

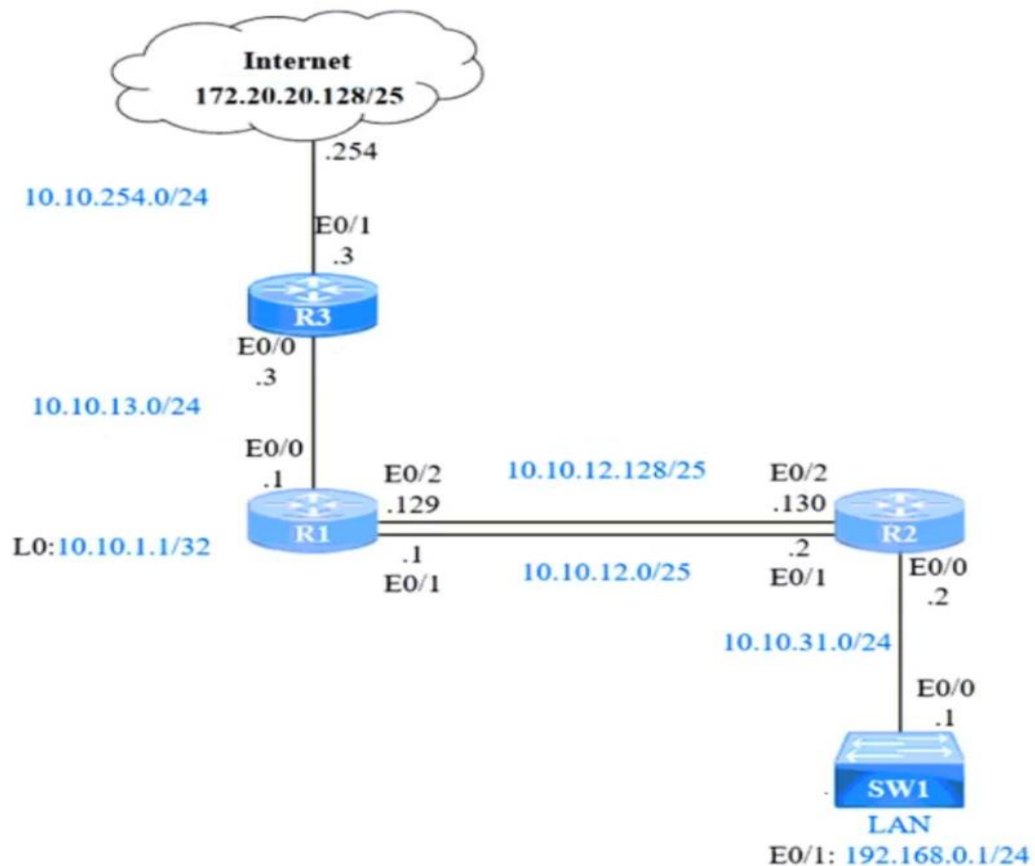
SW1(config)#ip dhcp snooping verify mac-address

SW1(config)#interface range gigabitEthernet 0/1-2

SW1(config-if-range)#ip dhcp snooping trust

Nhớ lưu cấu hình

Dạng 9:



Task:

IP connectivity and OSPF are preconfigured on all devices where necessary. Do not make any changes to the IP addressing or OSPF. The company

policy uses connected interfaces and next hops when configuring static routes except for load balancing or redundancy without tooting static.

Connectivity must be established between subnet 172.20.20.128/25 on the Internet and the LAN at 192.168.0.0/24 connected to SW1:

1. Configure reachability to the switch SW1 LAN subnet in router R2.

Task 1 R2(config)#ip route 192.168.0.0 255.255.255.0 10.10.31.1

2. Configure default reachability to the Internet subnet in router R1.

Task 2 R1(config)#ip route 0.0.0.0 0.0.0.0 10.10.13.3

3. Configure a single static route in router R2 to reach to the Internet subnet considering both redundant links between routers R1 and R2. A default route is NOT allowed in router R2.

#R2 (R2 đi ra internet)

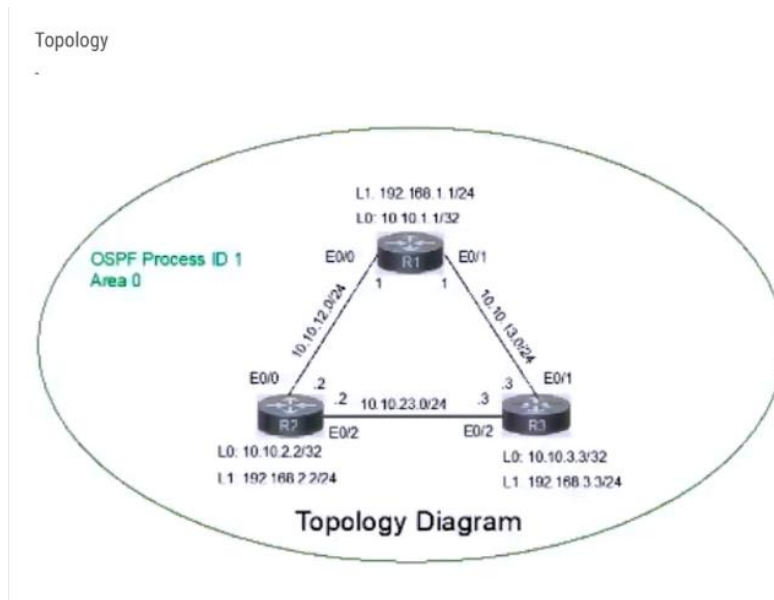
ip route 172.20.20.128 255.255.255.128 10.10.1.1 (Loopback)

Configure a static route in router R1 toward the switch SW1 LAN subnet where the primary link must be through Ethernet0/1, and the backup link must be through Ethernet0/2 using a floating route. Use the minimal administrative distance value when required.

R1 → Sw1

Nhớ lưu cấu hình

Dạng 10:



Tasks

IP connectivity between the three routers is configured. OSPF adjacencies must be established.

1. Configure R1 and R2 Router IDs using the interface IP addresses from the link that is shared between them. **(Route id is L0)**
2. Configure the R2 links with a max value facing R1 and R3. R2 must become the DR. R1 and R3 links facing R2 must remain with the default. OSPF configuration for DR election. Verify the configuration after clearing the OSPF process.

(Cấu hình cho 2 cổng của R2 : ip ospf priority 255, clear ip ospf process, yes)

3. Using a host wildcard mask, configure all three routers to advertise their respective Loopback1 networks. **(OSPF loop1) 0.0.0.0**
4. Configure the link between R1 and R3 to disable their ability to add other OSPF routers.

Task 4

R1(config)#router ospf 1

R1(config-router)#passive-interface gigabitEthernet 0/1

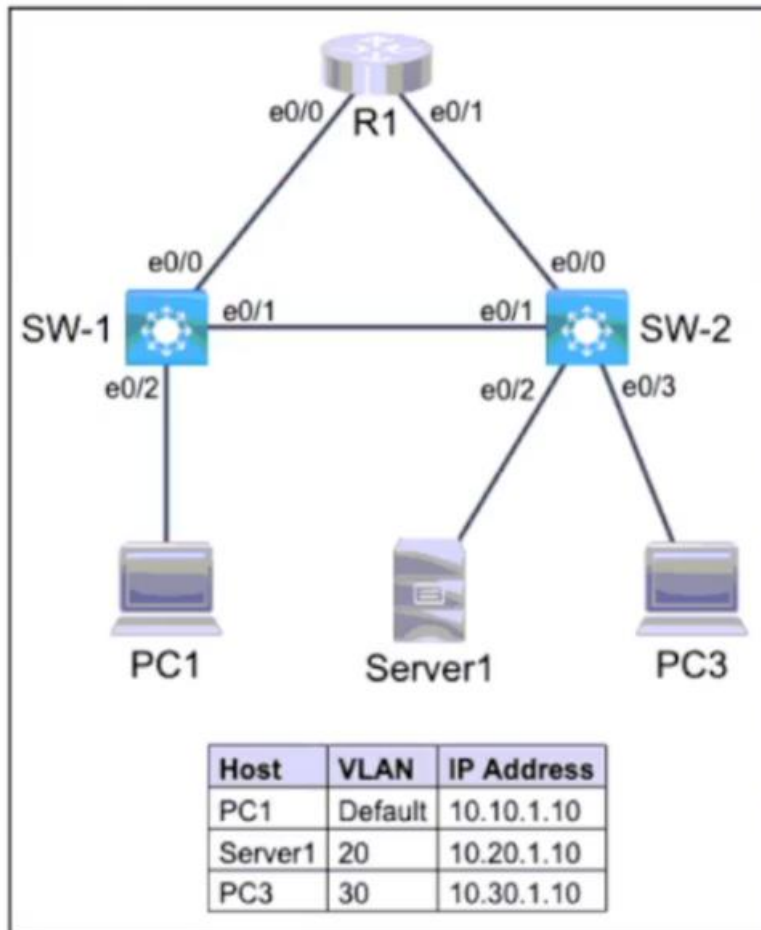
R3(config)#router ospf 1

R3(config-router)#passive-interface gigabitEthernet 0/1

```
R3(config)# interface G0/1
R3(config-if)# ip ospf network point-to-point
```

Chú ý lưu cấu hình

Dạng 11:



Tasks

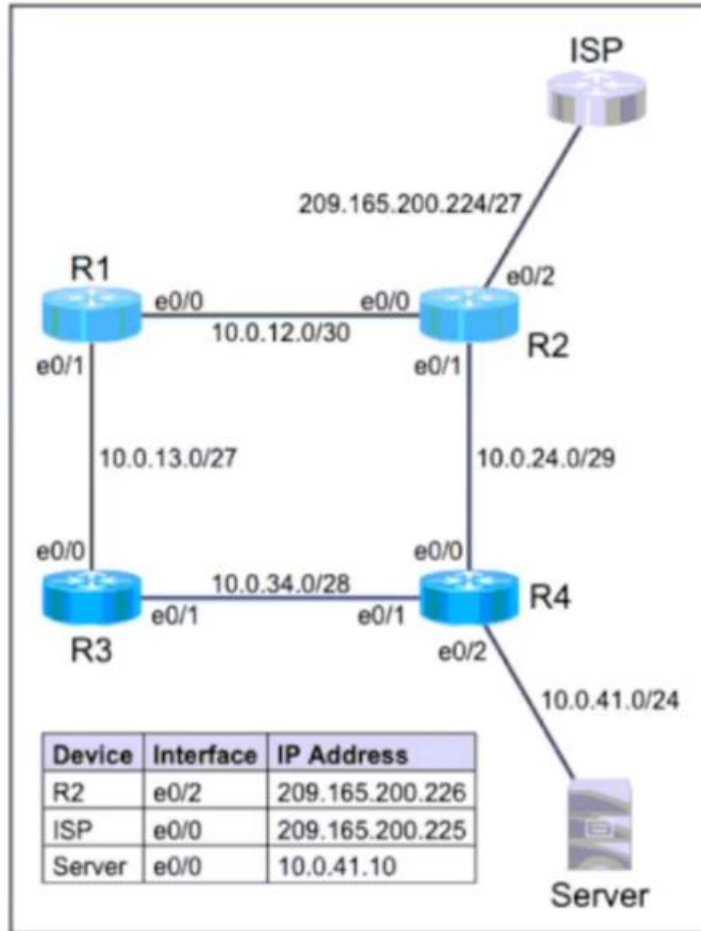
R1 has been pre-configured with all the necessary commands. All physical cabling is in place and verified. Connectivity from PC1, PC3, and the

Server must be established to the switches, and each port must only allow one VLAN.

1. Configure the VLAN connecting to the switch port for PC3 with the name "SALES"
2. Configure the switch port connecting to Server1
3. Configure the switch port connecting to PC3
4. Ensure R1 discovers SW-1 via the Cisco proprietary neighbor discovery protocol and all other devices on the network are unable to discover SW-1 (**cdp run , int e0/1- cdp enable, int e0/2 – no cdp enable**)

Nhớ lưu cấu hình

Dạng 12:



Tasks

All physical cabling is in place. Routers R3 and R4 are fully configured and inaccessible. Configure static routes for various connectivity to the ISP

and the LAN that resides on R4.

1. Configure a route on R1 to ensure that R1 prefers R2 when traffic is destined to the server only. **(R1→R2→Server)**
2. Configure a default route on R2 to the ISP
3. Configure a route on R1 to ensure that R1 will use R2 for the R4 LAN if the link fails between R3 and R4 (thêm 200 sau ip route)
4. Configure a route on R1 to ensure that R1 prefers R3 when traffic is destined to the R4 LAN at 10.0.41.0/24 (R1→R3→R4)

Task 1 R1(config)#ip route 10.0.41.10 255.255.255.255 gigabitEthernet 0/0

Task 2 R2(config)#ip route 0.0.0.0 0.0.0.0 209.162.200.225

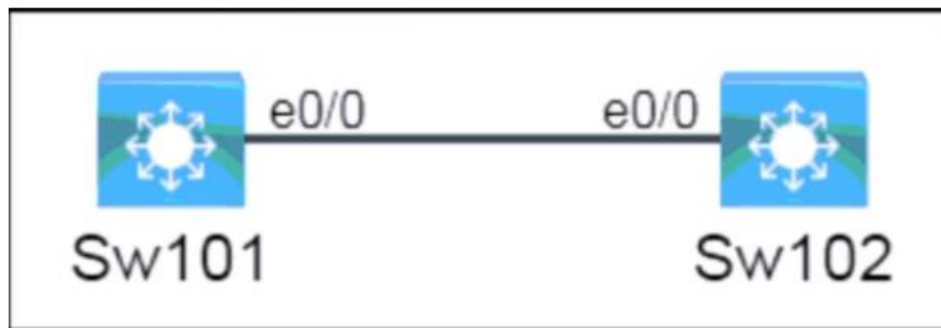
Task 3 R1(config)#ip route 10.0.41.0 255.255.255.0 gigabitEthernet 0/0 10

Task 4 R1(config)#ip route 10.0.41.0 255.255.255.0 gigabitEthernet 0/1

Nhớ lưu cấu hình

Dạng 13:

Topology



Tasks

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All physical cabling is in place. A company plans to deploy 16 new sites. The sites will utilize both IPv4 and IPv6 networks.

1. Subnet 172.16.0.0/16 to meet the subnet requirements and maximize the number of hosts
 - Using the second subnet
 - Assign the first usable IP address to e0/0 on Sw101
 - Assign the last usable IP address to e0/0 on Sw102
2. Subnet 2001:DB8::/50 to meet the subnet requirements and maximize the number of hosts
 - Using the second subnet
 - Assign an IPv6 GUA using a unique 64-Bit interface identifier on e0/0 on Sw101
 - Assign an IPv6 GUA using a unique 64-Bit interface identifier on e0/0 on Sw102

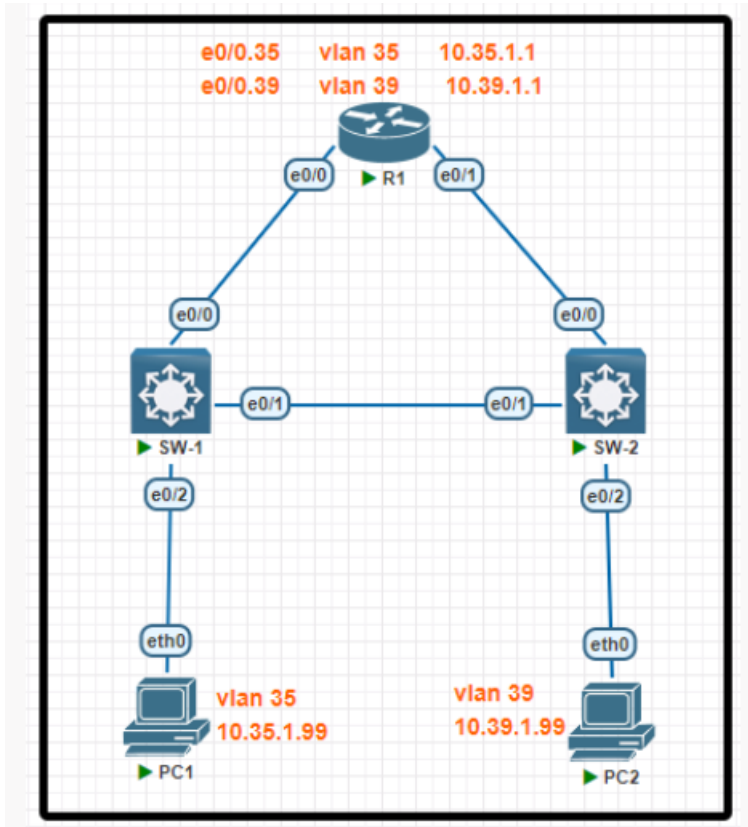
16 subnet $\rightarrow 2^x = 16 \rightarrow x=4$

First $\rightarrow 172.16.16.0 / 20$ (172.16.[00010000].[00000000] -> 172.16.16.1)

Last $\rightarrow 172.16.31.254/20$ (172.16.[00011111].[11111110] -> 172.16.31.254)

2001:DB8:0:400::/54 eui-64

Dạng 14:



Tasks

-

R1 has been pre-configured with all the necessary commands. All physical cabling is in place and verified. Connectivity for PC1 and PC2 must be established to the switches, and each port must only allow one VLAN.

1. Configure SW-1 with VLAN 35 and label it exactly as SALES
2. Configure SW-2 with VLAN 39 and label it exactly as MARKETING
3. Configure the switch port connecting to PC1
4. Configure the switch port connecting to PC2
5. Configure SW-1 and SW-2 for universal neighbor discovery using the industry standard protocol and disable it on the interface connecting to PC1

Task 5

SW-2(config)#lldp run

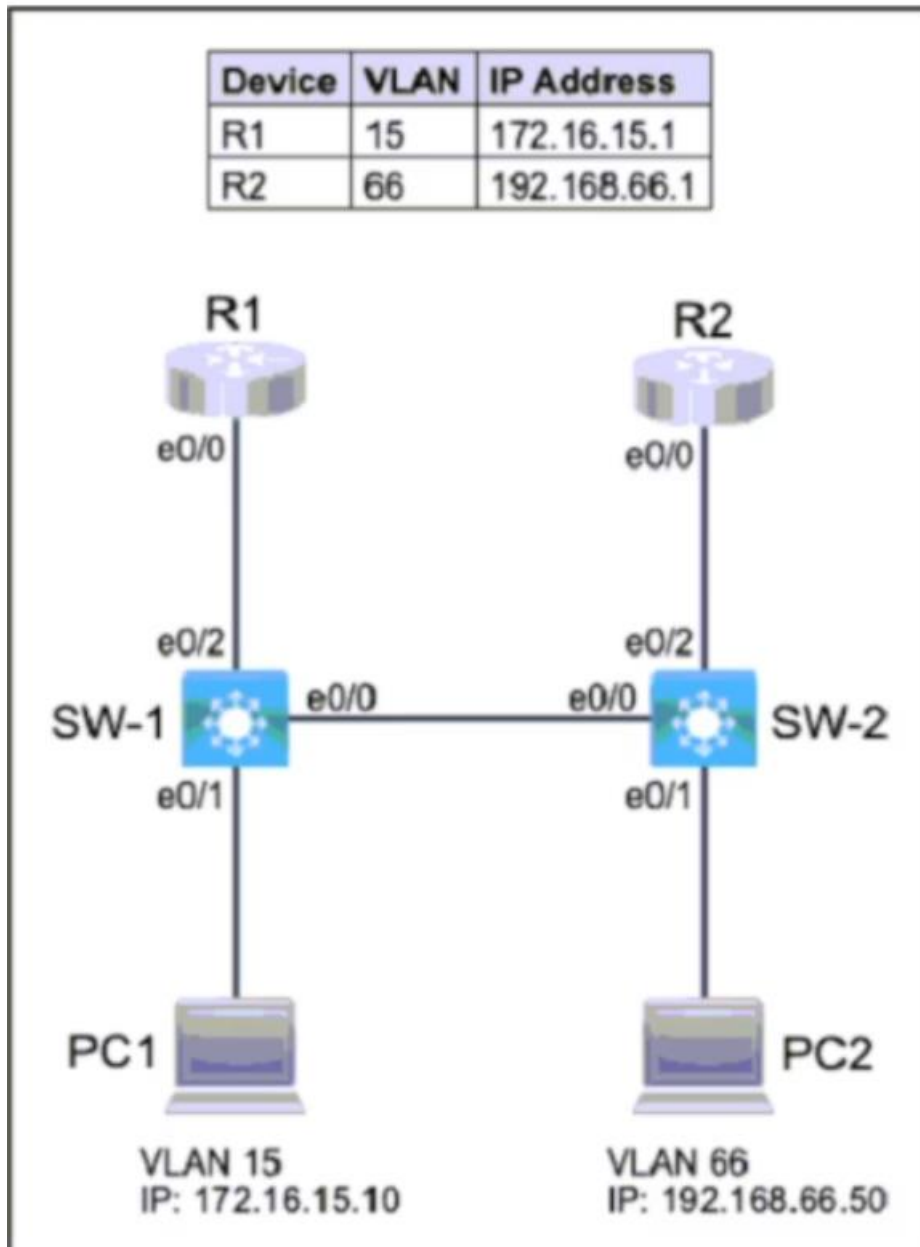
SW-1(config)#lldp run

SW-1(config)#interface gigabitEthernet 0/2

SW-1(config-if)#no lldp receive

SW-1(config-if)#no lldp transmit

Dạng 15:



Tasks

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R1 and R2 are pre-configured with all the necessary commands. All physical cabling is in place and verified. Connectivity for PC1 and PC2 must be established to the switches; each port must only allow one VLAN and be operational.

1. Configure SW-1 with VLAN 15 and label it exactly as OPS
2. Configure SW-2 with VLAN 66 and label it exactly as ENGINEERING
3. Configure the switch port connecting to PC1
4. Configure the switch port connecting to PC2
5. Configure the E0/2 connections on SW-1 and SW-2 for neighbor discovery using the vendor-neutral standard protocol and ensure that E0/0 on both switches uses the Cisco proprietary protocol

Task5 (Trên cả sw1 và sw2)

lldp run

cdp run

interface ethernet 0/2

lldp transmit

lldp receive

no cdp enable

interface ethernet 0/0

cdp enable

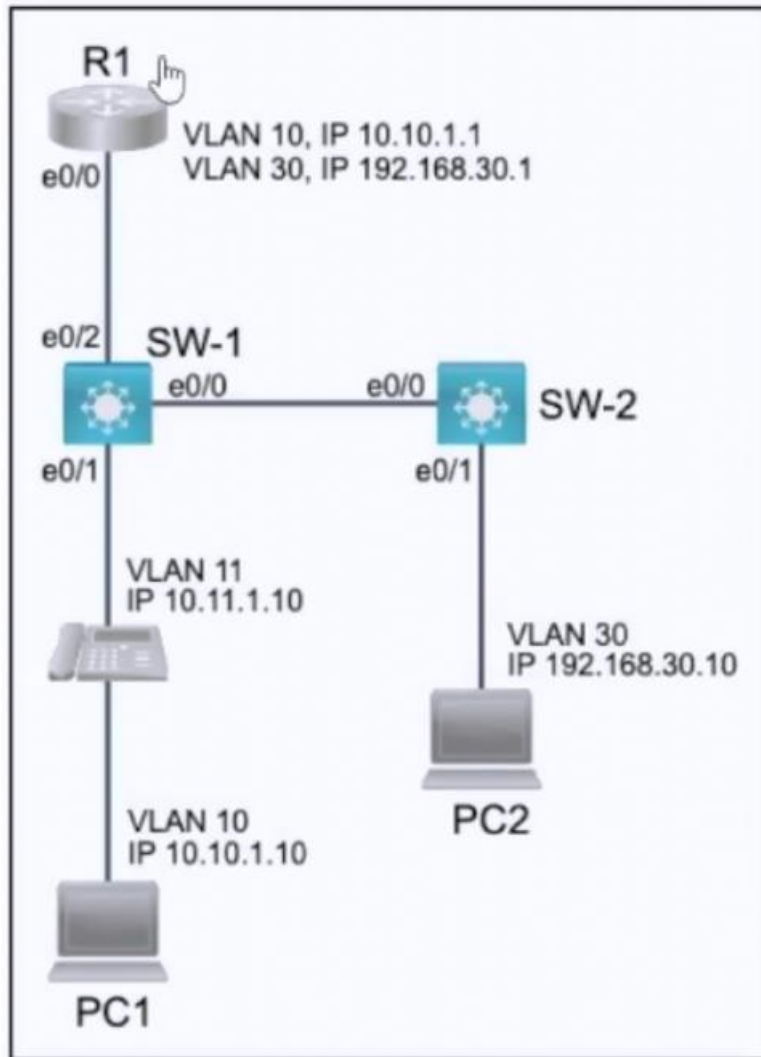
no lldp transmit

no lldp receive

switchport trunk encapsulation isl

switchport mode trunk

Dạng 16:



R1 has been pre-configured with all the necessary commands. All physical cabling is in place and verified. Connectivity to the end devices must be configured.

1. Configure SW-1 switch port 0/1 to carry traffic for the Cisco IP phone and PC
2. Configure SW-2 E0/1 to carry traffic for PC2
3. Configure VLAN 10 with the name "Engineering" on SW-1
4. Configure the link between SW-1 and SW-2 to use the vendor neutral neighbor discovery protocol
5. Configure the link on SW-1 to R1 so that it does not allow the Cisco neighbor discovery protocol to pass (**int e0/2 no cdp enable**)

Task 4: (SW1 tương tự SW2)

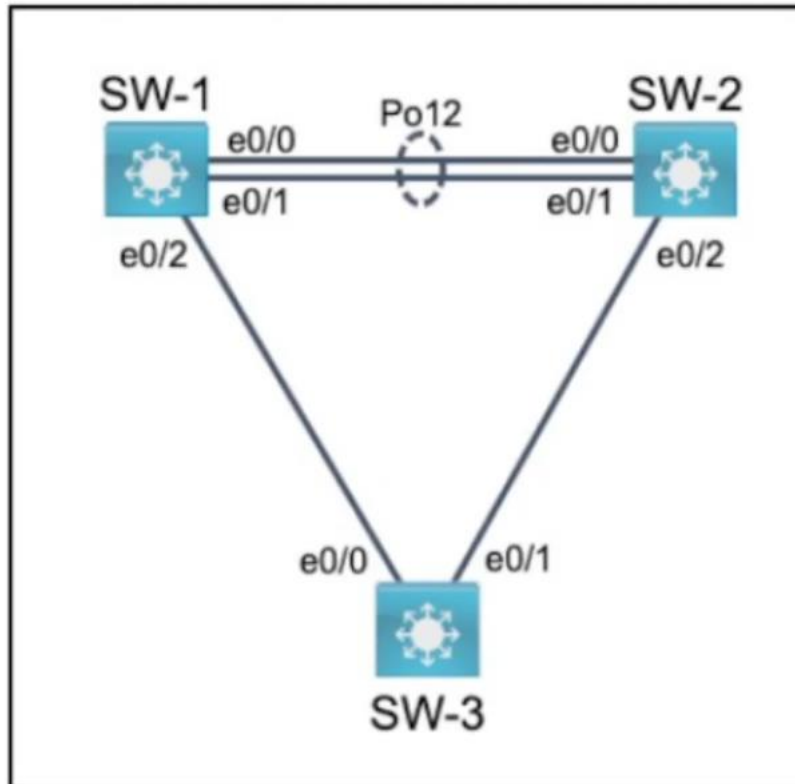
SW-1(Config)#lldp run

SW-1(Config)#int e0/0

SW-1(Config-if-range)#lldp transmit

SW-1(Config-if-range)#lldp receive

Dạng 17:



Tasks

VLANs 35 and 45 have been configured in all three switches. All physical connectivity has been installed and verified. All inter-switch links must be operational.

1. Configure SW-1 and SW-2 switch ports e0/0 and e0/1 for 802.1q trunking allowing all VLANs
2. Configure the inter-switch links on SW-1 e0/2, SW-2 e0/2, and SW-3 e0/0 and e0/1 to use native VLAN 35
3. Configure SW-1 and SW-2 switch ports e0/0 and e0/1 for link aggregation. SW-1 should immediately negotiate LACP and SW-2 must only respond to LACP requests

TASK #3

SW-1:

```
int range e0/0 - 1
```

```
channel-group 12 mode active
```

SW-2:

```
int range e0/0 - 1
```

```
channel-group 12 mode passive
```

```
exit
```

